Capture-Complete Learning for Transferable Bullet-Hell Control

Touhou Solver Research Project

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Abstract

We study control of the original Japanese Touhou 6 version 1.02h under Wine, with the scientific goal of completing a Lunatic no-miss, no-bomb route and eventually transferring the learning protocol to Touhou 8. The system is capture-complete but prediction-incomplete: it records coherent physical facts, filters only actions that intersect already-instantiated hazards over a fixed short horizon, and leaves action-relative future risk to a learned scorer. We separate the goal metric, no-miss completion probability, from the first optimization target, expected undiscounted physical HIT count over complete episodes. The latter is a dense surrogate and is not claimed to rank policies identically to the goal metric. The initial learner is deliberately limited to a causal decision-epoch dataset, a transparent supervised baseline, and immutable Wine evaluation. History encoders, offline value learning, uncertainty, adaptive collection, learned dynamics, and exact-prefix branching are admitted one at a time only after a simpler preregistered experiment exposes a concrete failure. The paper reports an audited six-stage infrastructure baseline with 108 HITs, a completed causal learner-view audit, a negative serial/parallel equivalence gate, a completed negative Stage 4 supervised-baseline gate, and an inconclusive optimization-only follow-up. A second optimization-only follow-up reached its train-only convergence criterion but failed the unchanged calibration gate. The fitted current-observation behavior clones improved held-out action log loss over action frequency but none passed all of its separately frozen promotion requirements; none was evaluated online. A read-only residual diagnosis rejected a global probability scale as sufficient and selected one small current-observation nonlinear ablation without inspecting scaled validation. That MLP converged and improved held-out log loss, accuracy, and Brier over the linear model, but failed the unchanged calibration gate and was not run online. A frozen target-contract diagnosis then verified the recorded propensities exactly and found neither sampled-label noise, 500 more optimization steps, nor replacing one-hot targets with full propensities sufficient. The residual is localized to the flat scorer's inductive bias for an exactly reconstructible action-conditioned lexicographic rule. The selected seven-parameter shared action scorer then exhausted its optimization timebox and was worse than the MLP on both splits, with zero final-tie agreement. This exposes a deeper mismatch between compensatory pointwise scores and the known non-compensatory collector rule; no learned policy was run online.

1 Motivation and Scope

Bullet-hell control combines dense continuous geometry, discrete input, delayed input pickup, partially observed pattern state, long episodes, and sparse but unambiguous failure events. A tempting engineering strategy is to combine a large object encoder, recurrent policy, offline value learner, ensemble uncertainty estimator, active collector, and learned world model at the outset. Such a stack cannot reveal whether a failure came from factual capture, causal action linkage, representation, value learning, exploration, or online inference.

Touhou-Offline instead adopts a *capture-complete, prediction-incomplete* boundary. The adapter records what physically exists at each coherent root, while learning is responsible for regularities not encoded by the minimal observed-hazard filter. The immediate engineering target is a complete Touhou 6 Lunatic route with HIT continuation and zero Bomb. The scientific target is Lunatic NMNB. Touhou 8 is reserved as a transfer test rather than a source of training-time identifiers.

This is a working protocol, not a result paper. In particular, we do not yet claim that the observed-hazard shield prevents all HITs, that object history is sufficient to infer unknown births, that offline RL improves the behavior policy, that a fixed corpus is sufficient, that exact-prefix branching is possible, that accelerated or parallel Wine is evidence-equivalent, or that the method transfers to Touhou 8.

1.1 Frozen outcome and non-goals

The required outcome is one immutable policy that reduces physical HIT count on complete normal-speed original-Wine routes without Bomb, infrastructure failure, forbidden actor input, or widening the observed-hazard action set. Every promoted result must also report NMNB completion rate. The first result need not complete NMNB; it must establish a correctly attributed improvement.

The initial baseline does not attempt online ECL interpretation, global safety proof, exact future-birth prediction, stage or spell routing, policy updates during a run, parallel collection, counterfactual simulation, or a novelty claim. These are non-goals unless a named experiment later earns one of them.

2 Objective and Causal Unit

For a complete physical episode τ, define

$$c_t = \mathbf{1}[\text{a physical life loss is observed on transition } t], \qquad C(\tau) = \sum_{t=0}^{T-1} c_t. \tag{1}$$

Bomb is forbidden and infrastructure failures are reported separately rather than converted into reward. Clearance, graze, score, power, rank, route progress, stage identity, and source identities are not reward.

On the declared evaluation population of complete, infrastructure-valid, zero-Bomb routes, the scientific goal metric is

$$J_{\text{NMNB}}(\pi) = \Pr_{\pi}[C(\tau) = 0]. \tag{2}$$

The first optimization target is

$$J_{\text{HIT}}(\pi) = \mathbb{E}_{\pi}[C(\tau)], \qquad \gamma = 1, \tag{3}$$

with terminal value zero only at the declared physical episode terminal. The two objectives are not interchangeable: equal expected HIT count can hide very different NMNB probabilities. Nevertheless, because C is a nonnegative integer,

$$1 - J_{\text{NMNB}}(\pi) = \Pr_{\pi}[C(\tau) \geq 1] \leq \mathbb{E}_{\pi}[C(\tau)] \tag{4}$$

on that population. HIT count therefore supplies a dense initial surrogate whose claim is exactly "fewer expected HITs," not "higher NMNB probability." Incomplete or invalid trials are never removed silently: their rates and causes are a separate promotion gate. A survival or distributional objective is a separate candidate experiment and may not be introduced silently.

The raw corpus remains frame-linked, but the policy intervention is not every raw frame. A command can remain in an input lease until Wine witnesses its pickup. A derived learner row must therefore join one policy invocation to the next or to the physical episode terminal and contain the published command, its complete behavior probability, witnessed sampled/executed actions, elapsed game frames, accumulated physical HIT cost, the next decision root or terminal, and all exclusion reasons. This decision-epoch view is a causal projection of factual Wine rows, not a simulated successor or a hand-designed gameplay option.

For reward-maximizing implementations the only task reward is $r_k = -C_k$, so maximizing $\mathbb{E}[\sum_k r_k]$ is exactly minimizing J_{HIT}; a cost-form implementation may minimize C_k directly. There is no discount, stage-boundary terminal, first-HIT terminal, or shaped progress term hidden in either convention.

Post-HIT roots remain in the complete episode and in the primary HIT-count view. Separate pre-first-HIT and time-since-HIT views may measure survival-conditioned coverage or train auxiliary factual predictors, but the initial task learner may not relabel the first HIT as the episode terminal.

3 Research Questions

RQ1: portable representation. After exact decision-epoch reconstruction, do current portable physical facts predict factual behavior and future HIT risk on held-out complete episodes, and does a short physical/action history improve them without privileged identity?

RQ2: minimal policy improvement. On one frozen representation and corpus, does the smallest justified offline value learner reduce complete-route expected HIT count relative to the behavior and behavior-cloning baselines under original Wine?

RQ3: coverage and interaction. If the fixed-corpus learner fails, is the demonstrated cause inadequate action/state support, survival-conditioned occupancy, representation, or value estimation; and does one frozen-policy batch of additional Wine data resolve that named failure more efficiently than uniform collection?

RQ4: transfer. Can the episode schema, causal learner view, feature contract, objective, exporter, and evaluator remain unchanged when only the Touhou 6 environment adapter and configuration are replaced by a Touhou 8 adapter?

The source-order question is no longer an open architecture choice. TH06 permits a hazard born after one paused root to enter collision processing before the next completed root. Linking and predicting this class in Wine remains an experiment, but it cannot justify an online birth envelope or ECL interpreter.

4 System Boundary

The system has three independently testable components.

1. A Wine adapter pauses the exact process and records a coherent physical root, input delivery, behavior probability, outcomes, lifecycle, and provenance.

2. A native shield filters actions whose bounded paths overlap hazards already instantiated at that root.
3. An offline learner consumes complete generic episodes and exports a frozen scorer. The scorer may rank the shield set but may never add an action to it.

There is no mandatory online ECL interpreter, stage route, spell rule, boss rule, frame window, coordinate target, or pattern exception. Extracted source scripts may support forensic analysis of the immediate-birth boundary, but source identities are not actor inputs and their successful decoding is not a corpus-admission condition. HIT is an observed cost and does not terminate data collection; Bomb is forbidden.

5 Minimal Online Shield

At decision root t, let x_t be player state, A the 18 directional/focus actions, and O_t the instantiated bullets, lasers, and enemy collision bodies. The adapter lowers O_t to per-frame hazard primitives for a fixed horizon $H = 4$. For each candidate $a \in A$, the native kernel advances the player under the measured movement constants and bounded input-delivery paths. It returns

$$A_{\text{shield}}(x_t, O_t) = \{a \in A : d(x^a_{t+k}, O_{t,k}) > m,\ k = 1, \ldots, H\}, \tag{5}$$

where $m = 0.35$ pixels is the fixed additional collision margin and d uses the source collision geometry.

The claim is deliberately narrow: a returned action is clear of the stored projection of hazards observed at root t. The shield makes no claim about a hazard born after t. The source-order audit at TH06 commit `cc475a0bc3fef38683b0f02224c87ddba0a021d9` establishes this boundary: Player update priority 7 precedes EnemyManager priority 9 and BulletManager priority 11; enemy execution may instantiate a hazard which BulletManager can test for collision in that same update. The audited definitions are in `src/ChainPriorities.hpp`, `src/EnemyManager.cpp`, and `src/BulletManager.cpp` of that source revision. Such births are ordinary partially observed learning outcomes, not justification for a title-specific online predictor. Empty shield sets are recorded as control dead ends; they are not silently widened.

The online path therefore contains only coherent capture, observed-object kinematics, native collision filtering, frozen-policy scoring, and input publication. A retired hand-written long-horizon beam planner is excluded so that gameplay changes can be attributed to the learned policy rather than an untracked rule system.

6 Algorithm-Independent Episode Data

An episode is an immutable sequence

$$\tau_i = \{(o_t, A^{\text{shield}}_t, a_t, \mu_t, o_{t+1}, c_t, b_t)\}_{t=0}^{T_i-1}, \tag{6}$$

where o_t contains factual physical state, a_t distinguishes proposed, published, sampled, and physically witnessed execution, $\mu_t(a)$ is the complete behavior distribution over the admissible actions, c_t is the physical HIT cost, and b_t records resource deltas, lifecycle boundaries, and infrastructure exclusions. Frames and transitions are joined by immutable references and shard hashes. A complete episode has exactly one fewer transition than frame.

The learner derives decision epochs k without changing the raw episode:

$$d_k = (h_k, u_k, \mu_k, \Delta_k, C_k, h_{k+1}, e_k). \tag{7}$$

Here h_k is a versioned portable history ending at a root where the immutable policy was invoked, u_k is the published command or $\perp$ when publication did not occur, Δ_k is elapsed game frames to the next invocation or the episode terminal, C_k is the sum of factual HIT indicators over that interval, and e_k contains publication, pickup, observation-gap, and learning-exclusion evidence. A $\perp$ row is retained for accounting but is not an action-conditioned training sample. Commanded, sampled, and executed actions remain distinct. The decision rows must conserve complete-episode HIT count exactly.

The canonical raw layer retains measured player state; occupied bullet, laser, enemy, player-shot, and item records; visual and collision geometry; resource counters; input state; clocks; capture retries; and executable, native, policy, and schema provenance. Derived tensors are versioned products rather than replacements for raw shards. This permits multiple feature builders and RL algorithms to consume identical episode identities.

The data contract has four distinct planes. The authoritative Wine plane is a dense title-specific capture; the factual-transition plane binds policy and input delivery to the next observed root; the portable plane exposes only actor-admissible physical observations; and derived features, histories, labels, and models are reproducible products. A recorded field is therefore not automatically an actor input, reward, or auxiliary target.

Current control roots retain subpixel player position and hitbox, movement constants and witnessed input; raw bullet-pool tails and visual mappings; raw laser, enemy, enemy-manager, and spawn-bullet records; decoded physical enemy bodies, lasers, items, and player attacks; resource and lifecycle counters; and capture integrity. Decision evidence retains the exact four-frame observed hazard primitives, collision margin, complete admissible mask, and each action's bounded clearance and final player position. Transition evidence keeps baseline, proposed, published, commanded, sampled, and executed actions distinct and stores the complete behavior distribution, factual successor, elapsed frames, HIT/Bomb/resource outcomes, and every exclusion reason.

Data plane	Representative retained facts	Authority and use
Wine root	Player/hitbox, raw and decoded instantiated objects, attacks, items, resources, lifecycle	Physical evidence, shield input, parity replay, and later bounded portable representations
Decision and input	Shield primitives and mask, policy identities, all action witnesses, complete propensities	Temporal/action alignment, BC targets, calibration, support, and later OPE
Transition outcome	Linked successor, elapsed frames, HIT, Bomb, deltas, boundaries, exclusions	Factual dynamics/risk labels and undiscounted physical HIT cost; never counterfactual successors
Integrity and provenance	Shard hashes, executable/native/policy hashes, schemas, latency, gaps, drops, completion and cleanup	Corpus admission and attribution; never gameplay features

Table 1: Separation of recorded facts from actor-visible and derived data. Recorder richness supports auditing and future falsifiable projections; it is not permission to expose privileged identity to the policy.

The first actor projection is deliberately much smaller: player position, power, bullet/laser and admissible-action counts, current action, and for each of 18 actions its shield legality, clearance-known bit, clearance, and final position. This gives 114 scalar features. Dense object facts may support one later bounded object-set experiment, but raw addresses, stage/boss/spell/ECL or RNG identity, run or absolute-frame identity, and future outcomes are forbidden actor inputs. Score, graze, clearance, and route progress are not reward; clearance is retained only as action-conditioned observation and parity evidence.

The same admitted episode may expose three non-authoritative derived views:

1. the complete decision-epoch HIT-cost view used by the initial task;
2. a pre-first-HIT or time-since-HIT diagnostic view used only to measure survival-conditioned coverage and factual prediction;
3. a complete physical view, including post-HIT roots, used for auxiliary representation and dynamics targets.

Every view names the same episode and raw hashes. No view may invent a next state, turn an unexecuted action into a transition, or admit a diagnostic stop-on-HIT run as training evidence.

Deployable features may contain normalized player state, unordered physical object primitives, action history, declared portable resources, and the shield mask. They may not contain stage, boss, spell, ECL, RNG, memory address, run identifier, future outcome, or post-treatment facts. Unordered hazard sets motivate permutation-invariant encoders such as Deep Sets [4], but this complexity is admitted only after a transparent vector baseline passes.

7 Minimal Learning Architecture

The strongest simple baseline has exactly three independently changeable parts:

1. a deterministic builder that converts complete factual episodes into audited decision rows and whole-episode splits;
2. a transparent current-observation behavior-cloning scorer,
$$\mathcal{L}_{\mathrm{BC}}(\theta) = -\sum_{i,k} \log \pi_\theta(u_k \mid h_k, A_k^{\mathrm{shield}}); \tag{8}$$
3. an immutable exporter that reproduces the offline action distribution bit-for-bit, ranks only observed-shield-admissible actions, and is evaluated on complete original-Wine episodes.

Action frequency and the existing reactive policy are non-fitted controls. BC is a plumbing and representation test, not a policy-improvement claim. The first known constraint is that the current online policy API exposes scalar player facts, object counts, and shield summaries but not the physical object history contemplated by RQ1. The baseline intentionally uses that smaller surface first. An API expansion is earned only after an offline held-out experiment shows that a precisely specified additional portable history improves a factual target and after offline/online feature parity is testable.

8 Candidate Ladder and Admission Gates

Only one rung may change at a time. Every rung freezes its corpus inventory, whole-episode split, feature schema, target schema, seeds, acceptance statistic, and artifact hashes before fitting.

L0: causal data audit (completed). Build and test decision-epoch rows. Admission requires exact raw linkage, command/sample/execution separation, propensity normalization, interval and episode HIT conservation, exclusion replay, forbidden-feature rejection, and two materially different feature builders reading the same episode identities. No model result is meaningful before L0 passes. The implementation and E0 replay evidence described in E2 satisfy this gate.

L1: transparent supervised baseline (completed negative). After L0, collect the initial training inventory serially, with unchanged retail clock/update semantics, coherent capture suspension, the frozen declared 20% uniform observed-shield mixture, and complete episodes; this is non-adaptive baseline collection, not L5. Freeze the episode count and split before fitting. Report per-admissible-mask action frequency, the reactive behavior baseline, and current-observation BC. BC passes the learnability test only if its held-out complete-episode negative log likelihood improves over action frequency with a preregistered episode-bootstrap interval and its calibration does not regress beyond a frozen tolerance. Accuracy alone is insufficient. E3 passed the likelihood condition but failed calibration, so L1 emitted no online candidate and does not admit L2 without a new transparent preregistration. Read-only replay subsequently bounded the first failure at optimization rather than capture: the portable roots reconstruct the recorded reactive choice exactly, train and validation show the same under-confidence, and the frozen train gradient is not stationary. L1b therefore changes only the train-only stopping rule and corrects the ECE implementation; it does not change data, split, representation, target, model class, regularization, or comparator. L1b exhausted its frozen update timebox before convergence and is inconclusive, so it still does not admit L2. L1c is the narrow continuation: it restarts the same fit from zero and changes only the maximum update ceiling from 2,000 to 10,000, retaining the same train-only 0.01 relative-gradient early stop and evaluating validation exactly once after stopping. No optimizer, hyperparameter, representation, comparator, or gate changes. L1c reached the gradient gate and failed calibration, rejecting this linear current-observation candidate under the frozen joint gate. A read-only residual decomposition may choose one subsequent representation ablation; L2 is not yet admitted. That diagnosis is frozen before execution: it may fit one positive global logit scale on training episodes only, but it does not evaluate the scaled validation distribution or fit a deployable model. If train calibration is repaired, it selects a scalar-calibration experiment; otherwise exact feature-only reactive reconstruction selects one small current-observation nonlinear model. History is not a candidate for this behavior target because the declared collector distribution is a current-root reactive choice plus an independent uniform draw.

L2: factual action and risk probes. First fit current-observation action-conditioned probes for executed $\Delta x/\Delta y$, observed-shield-set collapse, and physical HIT within fixed horizons. These diagnose command alignment and whether the captured current root contains learnable risk signal; they do not alter reward or make a gameplay claim. Add one short, fixed physical/action history only if a current-observation probe fails its separately frozen proper-score gate and the failure is attributable to temporal ambiguity. History is retained only if it then improves the same held-out target with an episode-level interval. Deep Sets [4] are considered only if a transparent bounded vector summary fails; a recurrent encoder is considered only if history has already shown value. Auxiliary heads remain separate from reward.

L3: one offline value method. Only after L1–L2 establish a usable representation may BC be compared with one expected-HIT value learner on the identical data. IQL is the first candidate because its critic avoids directly evaluating unseen actions and its actor remains advantage-weighted behavior cloning [2]. CQL is not run in parallel; it is admitted only if the IQL experiment identifies unsupported-action overestimation that its conservatism is designed to address [3]. Offline metrics can reject a candidate but cannot promote it without complete Wine evaluation.

L4: objective alternatives. A bounded survival or distributional HIT model is admitted only when the corpus contains enough late no-HIT prefixes or zero-HIT episodes to avoid a degenerate all-failure target, or when policies with similar expected HIT count show a measured difference in NMNB probability. Until then, first-HIT survival is a diagnostic, not the task terminal or deployment score.

L5: interaction and uncertainty. Frozen-policy batch collection is admitted only after L3 demonstrates either policy improvement or a measured support failure. The first additional batch comparison repeats the declared uniform-shield mixture at a fixed interaction budget. Ensemble uncertainty, active collection, OPE, or baseline bootstrapping may each be tested later against one named support or calibration failure. A positive action-probability floor does not by itself establish state coverage or reliable long-horizon OPE.

L6: learned dynamics and exact-prefix branching. Learned birth models or risk-aware MPC are considered only if model-free value learning fails while held-out future-hazard prediction succeeds. Predictions remain scores and never extend the shield certificate. Real-Wine branching is considered only after a serial same-seed, same-policy, exact-action-prefix gate reaches a target root with identical factual and input-delivery digests. The negative parallel gate does not establish this ability. Every training branch must still be a complete admitted Wine episode.

Decision Transformer [1], large recurrent world models, simultaneous IQL/CQL sweeps, and the proposed SABER-RL composite (survival critic, recurrent belief, ensemble uncertainty, active collection, and Wine branching) are not active candidates. SABER-RL is decomposed across L2 and L4–L6; neither the name nor the combination is a novelty claim. These methods add independent failure modes before the baseline has exposed a constraint they solve.

8.1 Algorithm and policy record

Policy or method	Status	Evidence or admission condition
Reactive scorer	Executed control	E0 completed one route with 108 HITs; this is infrastructure evidence, not a learned-policy result.
20% uniform-shield mixture	Executed collection	Twelve new serial Stage 4 episodes completed on attempt 1 with 139 factual HITs; this is a training inventory, not policy-improvement evidence.
Action frequency	Evaluated comparator	Validation NLL 2.286394 and ECE 0.008749 on the frozen four-episode split.
Current-observation BC	Negative held-out gate	Validation NLL 2.014976 beat action frequency, but ECE 0.124251 exceeded the frozen 0.028749 bound; no Wine canary ran.
Current-observation BC convergence follow-up	Inconclusive	Validation NLL improved to 1.720755 and ECE to 0.055148, but 2,000 updates ended at gradient ratio 0.0196 rather than the frozen 0.01 threshold; no canary ran.
Current-observation BC timebox extension	Negative held-out gate	The identical fit converged at update 4,631 and validation NLL improved to 1.691664, but ECE 0.059713 exceeded the unchanged 0.028732 limit; no canary ran.
L1c residual decomposition	Completed	Train-only scaling remained miscalibrated; current features reconstructed the collector exactly while the linear model reproduced only 66–67% of its reactive choices, selecting one small current-observation MLP.
Small current-observation MLP	Negative held-out gate	The fixed 114–32–18 model converged and improved validation NLL to 1.568672, but ECE 0.076007 exceeded the unchanged 0.028732 limit; no canary ran.
L1d target-contract diagnosis	Completed	All propensities were exact; paired train-only continuations rejected sampled-label noise, stopping, and discarded full propensities as sufficient explanations, selecting one structured current-observation scorer.
Shared action-conditioned BC	Inconclusive	Exhausted 10,000 updates at gradient ratio 0.011684; validation KL 0.986686 and reactive agreement 0.655371 were worse than L1d, with zero final-tie agreement.
Current-observation dynamics and factual risk probes	Not run	Next L2 diagnostics; history is admitted only by measured temporal ambiguity.
IQL	Not run	First L3 value candidate after representation learnability.
CQL	Not run	Admitted only after measured IQL unsupported-action overestimation.
Survival/distributional value	Not run	Admitted only by L4 coverage or objective evidence.
Deep Sets, recurrence, ensembles, active collection, learned MPC, branching	Not run	Each requires its distinct L2, L5, or L6 gate; none is active architecture.
SABER-RL composite	Not admitted	Its survival, recurrence, uncertainty, collection, and branching claims must each pass their separate rung before a combined-method experiment exists.

Table 2: Algorithm record at this checkpoint. “Not run” means no fit and no online learned candidate, not an unreported negative result.

9 Experimental Protocol

Every result must name the Git commit, executable hash, policy hash, schema, full command, complete episode IDs, and artifact paths. Splits are by complete episode. Negative results are retained. Infrastructure failures are reported separately and never counted as gameplay improvements. Before a fit, the experiment must additionally freeze its corpus inventory, split-group hashes, decision-row schema, feature and target schemas, seeds, comparator, acceptance statistic, and stopping rule. A post-hoc metric may be reported but cannot promote the candidate.

ID	Decisive question	Status
E0	Does the reset runner complete Practice 4–6 and one full route?	Completed
E1	Can a post-root birth hit before the next controllable root?	In progress
E2	Does the causal learner view preserve actions, propensities, and HITs?	Completed
E3	Do portable supervised baselines generalize by episode?	Negative result
E4	Does one offline value learner reduce complete-route HIT cost?	Not run
E5	Does shared-host parallel collection preserve exact facts?	Negative result
E6	Does a new frozen Wine batch resolve a measured coverage failure?	Not run
E7	Which surfaces change under a Touhou 8 adapter?	Not run

Table 3: Preregistered experiment ledger. Status changes require immutable artifacts.

9.1 E0: infrastructure baseline

Run natural Lunatic Practice Stages 4, 5, and 6 and then one natural six-stage route on original Wine with unchanged retail update semantics, coherent capture suspension, and the frozen baseline policy. HIT continuation and zero Bomb are mandatory. Accept only if every declared run completes, frames equal transitions plus one, no record is dropped, player successors and stored shield decisions replay exactly, every HIT has linked before/action/after evidence, and cleanup leaves no worker process. HIT count is measured, not gated.

At solver commit `af9900524520b72934a4c55e2f44118f88094633`, all three bounded Practice runs completed under isolated four-CPU workers after the atomic-input cleanup. Table 4 reports their complete audits. Every declared frame and transition loaded, every Bomb/callback/integrity/scope count was zero, all checked player successors were bit-exact, and every stored dense shield sample reproduced with no unsafe or conservative divergence. The higher Practice observation-gap rates motivated eight CPUs for the canonical route and future evidence collection; gap transitions remain explicit and are excluded rather than fabricated.

Stage	Frames	Transitions	HIT	Gap %	Successors	Shield
4	25,564	25,563	18	0.536	24,344	58
5	23,129	23,128	23	0.968	19,503	64
6	30,981	30,980	34	0.507	28,763	64

Table 4: Canonical E0 Practice evidence. Successor and shield columns are checked counts; every mismatch/divergence count is zero.

The Stage 4–6 run IDs are, respectively, `20260821T080408Z-471737400`, `20260821T081924Z-372154700`, and `20260821T080532Z-100425500`; their audit SHA-256 digests are `c4178c3c9e5b7d88f7a950d8fe18403630a8524320ae72dc0d0ce97c0a26bb2d`, `2677f2b9689c42c5483a2e278d077a`

`6b721623430d89f6c39a41bc92b3c34ebd`, and `e34bec2844dc7ce92f572326dd262a1e7f718d6798bc9a6c9dd7124692308c17`.

The natural-RNG full route then reached Ending with run ID `20260821T083303Z-499458600`: 135,561 frames, 135,560 transitions, 108 physical HITs, zero Bomb, zero dropped record, and zero capture, corpus, policy, trace, or other infrastructure failure. All six Stages were present. The audit classified all 108 HITs as observed-shield-empty, reproduced 64 dense shield samples without divergence, and matched 126,939 eligible player successors bit-for-bit. Only 28 links were excluded for observation gaps (0.0207%); capture and solve p99 were 8.044 and 0.918 ms respectively. The independent baseline verifier passed every declared check. The immutable SHA-256 digests for report, audit, run metadata, and manifest are respectively `f2577d31ca5a51108911b4d60a957a7e2e891edda5ba337fbca1f70386e37686`, `8d13d3196261f4f8710d8187cdec898e364b67f024696a8cca1991aff467baa9`, `9f9fbdd75dbe89651c789dc502dab6a50323a93ca4880517205383080 79e3396`, and `8423bae1ef93decd50605a5e47900ee0ded3a6096d6067fb4c0a8eca944ec1d8`. E0 is therefore complete; the 108-HIT value is a baseline measurement, not a policy-quality claim.

9.2 E1: immediate-birth boundary

Use source update order and adjacent physical roots around every HIT and newly observed overlapping object. Source order already proves that post-root birth and same-update collision are possible. The remaining experiment links this class to adjacent Wine roots and measures whether short physical history predicts its action-relative cost. No source/ECL envelope is added to the online shield; pattern-specific exceptions are forbidden.

9.3 E2: causal learner view

The existing generic loader already admits complete E0 episodes and enforces frame/transition linkage. E2 is not complete until it also removes optional source/ECL material, derives decision epochs from policy invocations and input leases, reproduces proposed/published/sampled/executed action sequences, conserves interval and episode HIT totals, validates complete propensities and exclusions, and lets two different feature builders consume the same raw episode identities. A feature-parity fixture must prove that the online exporter receives exactly the versioned values produced by offline replay. E2 changes no policy and collects no new gameplay evidence. At commit `2818861f4079bebfbd8443638ed0cb34236bd5e0`, the decision builder, two-path offline/online feature-parity fixture, masked linear BC fixture, and immutable export/reload tests passed as part of 176 repository tests. Read-only replay of canonical E0 Practice Stage 4 produced 24,385 decision epochs, retained 24,249 as action-conditioned learner rows, and conserved all 18 physical HITs; 7 HITs remained explicitly attached to excluded intervals. The canonical complete route produced 126,468 decision epochs, retained 126,424, and conserved all 108 HITs; one HIT remained in an excluded interval. Raw-frame and compact-context actions and propensities are checked exactly, forbidden source facts are absent from the portable root, and offline and online feature vectors are bit-exact. E2 is therefore complete.

9.4 E3: supervised representation baselines

After E2, collect the initial inventory serially with unchanged retail clock/update semantics, coherent capture suspension, and the frozen 20% uniform observed-shield mixture; its episode count, seeds, and stopping rule are declared before the first run. Then freeze its whole-episode train/validation split and compare the per-admissible-mask action-frequency control with current-observation BC. Only if the preregistered held-out negative-log-likelihood and calibration gate passes may one short fixed history be added. The same split may then support simple executed-action factual probes for HIT within fixed horizons and future observed-shield collapse. These probes test information content; they do not define reward, infer unexecuted successors, or constitute policy improvement.

The first E3 inventory is frozen in `experiments/l1-stage4-bc-v1.json`. It contains exactly 12 complete natural-RNG Lunatic Practice Stage 4 episodes under the 20% uniform observed-shield mixture, with independent preregistered policy RNG streams. Episode indices 2, 5, 8, and 11 are validation; the remaining eight are training. Collection uses one isolated eight-CPU Wine worker at a time because E5 did not admit parallel Wine. A physical HIT never causes retry. Bomb, incomplete storage, infrastructure failure, semantic/parity failure, capture p99 above 40 ms, solve p99 above 16.7 ms, observation-gap rate above 0.5%, or nonzero stale-retry rate rejects that attempt; each fixed policy slot permits at most three infrastructure attempts, after which the experiment stops incomplete without fitting.

The current-observation masked linear softmax fit is frozen at 100 full-batch epochs, learning rate 0.05, $L_2 = 10^{-4}$, seed 0, 2,000 episode-bootstrap draws, and at most 400,000 rows per split. BC passes only when the upper end of the 95% episode-bootstrap interval for validation NLL minus the per-mask Laplace action-frequency NLL is below zero and ten-bin ECE is no more than 0.02 worse than the comparator. No auxiliary target is active in L1. A passing artifact earns one complete natural-RNG Stage 4 online integration canary; that canary is excluded from training and makes no HIT-reduction claim.

At solver commit `559572a8f3699c06eb41080e6061e579a1156c33`, all 12 slots completed and passed admission on attempt 1 with no Bomb, rejection, retry, stale input, drop, semantic/parity failure, or leftover Wine worker. The inventory contains 296,667 authoritative roots, 296,655 transitions, 794 events, 287,545 decision epochs, 287,493 eligible rows, 139 conserved HITs, and 643 observed-shield control dead ends. All HITs remained in eligible decision intervals. Capture p99 ranged from 8.018 to 8.421 ms and solve p99 from 0.764 to 0.897 ms.

The manifest-declared corpus occupies 4,583,135,878 compressed bytes for 36,491,160,547 canonical uncompressed bytes, a 7.962-fold ratio. A post-hoc, non-promotional codec diagnostic streamed one fixed 157,969,806-byte frame shard: current gzip-3 used 17,178,795 bytes, gzip-9 14,779,903, zstd-3 7,327,695, and zstd-9 5,871,707. No shard was changed. L1 therefore retains the existing format; a future codec migration requires a versioned manifest and content-equivalence audit, while a compact derived decision cache is admitted only if repeated fitting is measured to be decode-bound.

The fit used 193,331 training rows and 94,162 validation rows. Validation BC NLL was 2.014976 versus 2.286394 for action frequency; the 2,000-draw episode-bootstrap interval for their difference was $[-0.287520, -0.254559]$, so the likelihood criterion passed. Validation ten-bin ECE was 0.124251, while the frozen comparator-plus-tolerance bound was $0.008749 + 0.02 = 0.028749$. Calibration therefore failed and the joint gate rejected policy `linear-behavior-clone-v1:cdb68b242791738d`. The result artifact decision is `stop-l1-bc-learnability`; no Wine canary, HIT comparison, or NMNB claim exists. The collection ledger, model, and result SHA-256 digests are respectively `614e44b1eab047041337a5ebcee01af3ef672d01a265d856d0dd58c530dc4c87`, `7b47fee4c261eb8dbde77e3e13391a49cc5771d27e9c4d3d546f6abb695cd245`, and `75810a82c9bf3ec0a1063b0d53e6814243afc08b3319268800a2eaff5ef5bae3`.

The result establishes learnable current-observation signal relative to action frequency but not an admissible behavior distribution. A read-only reliability decomposition may diagnose the frozen artifact; any refit, temperature calibration, optimizer change, or representation change is a new preregistered experiment. E4 and online learned-policy evaluation remain blocked.

That read-only decomposition reproduced the BC metrics and found systematically low confidence rather than a validation-only failure. Validation accuracy was 0.461290 while mean top confidence was 0.337672; train showed 0.468140 versus 0.341038. Each validation episode had ECE between 0.1151 and 0.1362, only 0.808% of validation rows used an unseen training shield mask, and the largest feature-mean displacement was 0.214 training SD. The frozen train gradient L2 norm was 0.197788, 19.1% of its initial 1.037370, while the train-NLL derivative along a uniform logit-scale increase was −0.271893. Thus the 100 full-batch updates did not reach a stationary train solution.

The permitted portable facts independently reconstructed the recorded reactive baseline on all 94,162 validation rows, excluding missing capture and the future-birth boundary as explanations for this action-imitation result. The linear BC reproduced that baseline on only 56.19% of rows. This may still expose linear expressivity after convergence: 58.3% of rows tied at maximum clearance and 43.4% had all clearances infinite, so the controller frequently used later tie breakers including nonlinear boundary reserve. That hypothesis is not admitted until optimization is resolved.

The decomposition also found that floating ECE edges failed to assign exact confidence 0.6 to any bin. Integer equal-width binning changes the frozen action-frequency ECE from 0.008749 to 0.008732 and leaves BC ECE at 0.124251; the corrected bound is 0.028732, so the historical L1 decision remains negative. The original result is not rewritten.

The separately frozen L1b protocol is `experiments/l1b-stage4-bc-convergence-v1.json`. It reuses the exact 12-episode inventory and eight/four split, initializes the identical masked linear softmax at zero, retains learning rate 0.05 and $L_2 = 10^{-4}$, and uses the corrected versioned ECE. After at least 100 updates, training stops when gradient L2 is at most 0.01 of its initialization value, or after 2,000 updates. The latter outcome is inconclusive and cannot be attributed to representation. Validation is evaluated exactly once after the train-only stop. The likelihood and comparator-plus-0.02 calibration conditions are unchanged. L1b collects no data and runs no Wine canary; a joint pass only admits a separately executed Stage 4 integration canary.

L1b ran at solver commit `7285ee76fe36eda1470844b3635eaddd64292d23`. It exhausted all 2,000 updates with final train gradient L2 0.020297, or 0.019566 of the initial 1.037370, and therefore missed the required 0.01 ratio. The optimization gate failed and the preregistered result is `inconclusive-l1b-optimization-not-converged`; this is not evidence that the linear representation failed.

Optimization nevertheless moved every descriptive held-out metric in the expected direction. Validation NLL fell from the L1 value 2.014976 to 1.720755, Brier from 0.728516 to 0.647767, ECE from 0.124251 to 0.055148, and accuracy rose from 0.461290 to 0.528111. The corrected action-frequency NLL and ECE were 2.286394 and 0.008732; the episode-bootstrap NLL-difference interval was $[-0.578257, -0.552904]$. The proper-score gate passed, but ECE remained above the corrected 0.028732 bound, so calibration also failed. Train NLL and ECE were 1.707485 and 0.054720, close to validation. No online canary ran and no HIT or NMNB claim exists. The plan, model, and result SHA-256 digests are respectively `d972fbda8b26cf37fe69438fc81423714c92f652652ba1667e5c27830bcdb592`, `22254afd55d97a740889e76c44617fd408d3354566a550abc3f74feb9b497ae5`, and `0d814a6f20036175e16a507256e79c3195ce4b48193b73c3fc51b7cc33d1d336`. Any convergence-suitable optimizer or larger timebox would be another frozen L1 experiment; validation may not select it.

The L1c protocol is frozen in `experiments/l1c-stage4-bc-timebox-v1.json`. It reuses the exact L1 corpus, whole-episode split, code, from-zero initialization, features, target, linear model, full-batch optimizer, learning rate, regularization, seed, comparators, and acceptance gates. Its single experimental change is a maximum of 10,000 rather than 2,000 updates. After at least 100 updates it still stops immediately when train gradient L2 is at most 0.01 of the same initial value; validation is evaluated once only after that train-only stop. The ceiling is five times L1b's optimization budget because L1b ended less than a factor of two above the frozen gradient threshold. This is a resource bound, not a validation-selected update. L1c collects no Wine data and runs no canary. If it converges but fails calibration, the transparent linear current-observation representation is rejected by the unchanged gate. If it still does not converge, the next smallest experiment is a separately frozen deterministic convergence-suitable optimizer, not history or a larger model. L1c ran at solver commit `171d92b95c55331ab23bec73226b6b81de6f64cf`. The train-only stop fired at update 4,631 with final gradient L2 0.010372, or 0.009998 of the initial 1.037370. The optimization gate therefore passed. Validation NLL was 1.691664 and the episode-bootstrap BC-minus-frequency interval was $[-0.607988, -0.581455]$, so the proper-score gate also passed. Validation ECE was 0.059713 against the unchanged comparator-plus-tolerance limit $0.008732 + 0.02 = 0.028732$; calibration and the joint gate failed. Validation accuracy and Brier were 0.543234 and 0.634249. Continued optimization from L1b improved NLL, Brier, and accuracy but worsened ECE by 0.004565. The preregistered decision is `stop-l1c-linear-current-observation`; policy `linear-behavior-clone-v1:368d286c449c3d7d` was not run online.

The immutable plan, model, and result SHA-256 digests are respectively `f1eef124b1717cac7f582a50964e793ed98a75e34ea41930e66a6f734d4cb8aa`, `b7e350ff86103ee803a6c2a856f5509b1a29b5080c02e78e5825de9a9cca16b9`, and `fac6fd06cd57f93ed276a6c5afa5a3c6c8da9bf8dd42935fac6b876ffe3dc1a1`. L1c rules out insufficient optimization time as a sufficient explanation for the calibration failure. It rejects the current feature/model/exported- probability combination under the declared gate, not BC in general or every model using current physical observations. It also does not distinguish nonlinear current-observation expressivity from missing temporal information. One read-only residual decomposition and then one separately frozen representation ablation are admissible; continued optimization of the same fit, an online canary, combined history/architecture changes, and IQL are not.

The residual protocol is frozen in `experiments/l1c-stage4-residual-diagnosis-v1.json`. It binds the L1c model and result, replays the same eight/four whole-episode split, and first reproduces all frozen metrics. A one-dimensional positive logit scale is fit by training NLL only. Branch selection uses its training NLL and ECE plus an exact reconstruction of the reactive baseline from the 114 current features; the scaled validation distribution remains unread. A successful train-only scale selects one separately preregistered scalar-calibration validation. Otherwise, exact current-feature reconstruction selects one small nonlinear current-observation model. Failure of the reconstruction invariant stops the ladder rather than silently selecting history.

The diagnosis ran at solver commit `a4e8232fbae5e3c4df9ecb8d7a787c5138d54e32`. The positive train-NLL optimum was logit scale 1.032470 (temperature 0.968551) and changed no argmax. It improved train NLL only from 1.679447 to 1.678613 and train ECE from 0.058741 to 0.046387, still above the frozen 0.028732 limit. The scaled validation distribution was not evaluated. Current features reconstructed the reactive baseline on every one of 193,331 train and 94,162 validation rows, ruling out temporal ambiguity for the collector target; its remaining 20% choice is an independent declared uniform draw. The linear model agreed with the reactive action on only 67.05% of train and 66.40% of validation rows. The exact collector oracle had train/validation NLL 1.004524/1.006665 and validation ECE 0.000756, so this gap is not irreducible exploration noise. The preregistered selection is therefore one small current-observation MLP, with temperature, history, auxiliary targets, and value learning excluded.

The immutable plan and diagnosis SHA-256 digests are respectively `a05cd1f70f2e689eb5018cf3393dd30c3681006ae7cf84b834cc5193b8e3bbab` and `e3c25c844c9ec6fdd47388fb3a9690375dcc77c2c7963d30ebfa180eaccce8ce`.

The selected L1d representation ablation is frozen before fitting in `experiments/l1d-stage4-bc-mlp-v1.json`. It reuses the immutable 12 episodes, eight/four whole-episode split, 114 current-observation features, shield mask, published executed-action target, learning rate 0.05, L2 10^{-4}, full-batch gradient descent, seed 0, 10,000-update ceiling, and the same train-only 0.01 relative-gradient stop. Its sole representational change is a fixed 114–32–18 network with one ReLU hidden layer. All-zero weights would leave the hidden units symmetric, so the one necessary companion change is fixed-seed He-normal weights with zero biases; initialization, width, and depth are not swept. There is no history, temperature, auxiliary target, privileged collector action, value loss, or new Wine data.

L1d's primary proper-score test is deliberately harder than the original L1 frequency comparison: the upper endpoint of a 2,000-draw complete-episode bootstrap interval for MLP-minus-frozen-L1c validation NLL must be below zero. It must also converge under the unchanged train-only criterion and keep validation ECE no greater than action-frequency ECE plus 0.02. Improvement over action frequency remains report-only and cannot replace direct evidence against the converged linear model. Validation is evaluated once after the train-only stop. Non-convergence is inconclusive; convergence followed by a failed direct-NLL or calibration gate rejects this small current-observation MLP. This offline fit runs no Wine canary and makes no HIT, value-learning, or NMNB claim. Only the joint supervised gate may admit a separately preregistered Stage 4 integration canary.

L1d ran at solver commit `c96845952572f41f4c263706fe76548b6927df7e`. Its train-only stop fired at update 3,121 with final gradient L2 0.020971, or 0.009996 of its fixed-seed initial value 2.098068, so optimization passed. Validation NLL was 1.568672, compared with 1.691664 for frozen converged L1c; the complete-episode MLP-minus-L1c bootstrap interval was $[-0.127334, -0.115239]$, so the direct representation gate passed. Validation accuracy and Brier also improved from 0.543234 and 0.634249 to 0.610076 and 0.572805. The report-only MLP-minus-frequency NLL interval was $[-0.734694, -0.700380]$.

Calibration nevertheless failed. Validation ECE was 0.076007 against the unchanged limit $0.008732 + 0.02 = 0.028732$; train ECE was similarly high at 0.080346. A read-only post-result reliability check found mean validation top confidence 0.549501 versus accuracy 0.610076. Most 0.2–0.7 confidence bins were under-confident, while the 0.8–0.9 bins were over-confident; no probability transform was fit or evaluated. The immutable decision is `stop-l1d-small-current-observation-mlp`. The plan, model, and result SHA-256 digests are respectively `ef1ae999fe75d434cd60f9126d50a2d7e89210f88e925626174e117a0dc7e8c5`, `3f8d54f2574e4830c0b2cc81c516d70bcc5637967600a8df8941860a8ddaa286`, and `aaf62b7e6a696c4d72e7f61a8b68e772d45e6f320b23c7e963b84b15d7657bf2`. Thus the nonlinear current-observation hypothesis receives partial support on proper score, accuracy, and Brier, but this exact exported probability model fails the joint supervised gate. It runs no Wine canary and admits neither history nor value learning. Any calibration diagnosis or further ablation must be separately frozen; this result cannot be repaired post hoc.

The root-cause protocol is frozen in `experiments/l1d-stage4-target-diagnosis-v1.json`. It first audits every eligible row against the declared 0.8 reactive plus 0.2 uniform-shield distribution and measures frozen L1d against both sampled actions and the recorded full distribution. It then starts two non-deployable branches from the exact frozen L1d state: 500 additional full-batch train-only updates with the historical one-hot published-action objective, and 500 with cross-entropy to the recorded full propensities. Both retain learning rate 0.05 and L2 10^{-4} and report fixed checkpoints 0, 100, 250, and 500. Branch parameters are never serialized and neither branch is evaluated on validation.

The diagnosis attributes a recorder/dataset defect only if a recorded mixture differs by more than 10^{-15}. Otherwise a final soft-target train KL advantage of at least 0.005 nat selects one separately preregistered full-propensity target correction. Failing that, a hard-target KL reduction of at least 0.01 nat selects one stopping correction. If neither occurs while a large distribution and reactive-action residual remains, the next candidate is one structured current-observation scorer, not history. This diagnosis runs no Wine, fits no deployable model, does not revise the negative

L1d result, and cannot admit value learning.

The diagnosis ran at solver commit `b12efebd995832fd68669af163255419cd517dc7`. All 193,331 train and 94,162 validation rows matched the declared behavior distribution with maximum absolute error exactly zero. Frozen L1d validation $\mathrm{KL}(\mu\|\pi)$ was 0.563604 and exact-distribution top-label calibration error was 0.077039, close to sampled-action ECE 0.076007; sampling noise is therefore not a sufficient explanation. The current features reconstructed the reactive action on every row, whereas L1d agreed on only 0.754633 of train and 0.748550 of validation rows. On validation, agreement was 0.801009 when boundary reserve was decisive and 0.725546 when clearance was decisive, but only 0.003086 for focused and 0.039474 for lexical final tie breaks.

After 500 additional hard-target train updates, train $\mathrm{KL}(\mu\|\pi)$ fell from 0.552480 to 0.542529, a 0.009951 reduction just below the frozen 0.01 materiality threshold. The paired full-propensity branch ended at 0.541191, only 0.001338 nat better than the hard branch and below the frozen 0.005 threshold. Thus the result does not claim that further optimization or exact soft labels have zero effect; it rejects either as sufficient for the large residual under the preregistered bounds. The selected attribution is a flat-model inductive-bias mismatch for the known action-conditioned lexicographic relation. History remains unsupported. The experiment-plan and result SHA-256 digests are `eabe1743d40bc1cb8d5ab949826658ac46e483ef564ce5e922a4f84739f05738` and `6aaee7631249dc30f0ac60fa76a3e72a2c42ab5bda580340ca6b14722b056e4f`, respectively.

The diagnosis also exposes two prospective protocol corrections without changing any historical decision. First, reducing a captured full behavior distribution to one sampled action discards exact supervision, although its measured effect here was only 0.001338 nat. Second, top-label ECE relative to a diffuse action-frequency predictor is not a proper distributional comparison: such a predictor can have small ECE despite poor likelihood. A future behavior-model gate should compare episode-bootstrap cross-entropy or $\mathrm{KL}(\mu\|\pi)$ directly against the recorded full distribution. L1d remains negative because it is also poor under that exact target; this is not a post-hoc revision of its frozen gate.

The selected L1e test is frozen in `experiments/l1e-stage4-bc-shared-action-v1.json`. It retains the same 12 episodes, eight/four split, published executed-action target, observed-shield mask, learning rate 0.05, L2 10^{-4}, all-zero initialization, 10,000-update ceiling, and train-only 0.01 relative-gradient stop. Its only learner change is one seven-parameter linear score shared across all 18 actions. For each action it reads clearance unknownness and value, boundary reserve derived from the already captured final coordinates, equality to the current action, stationary and focused action attributes, and fixed lexical action-name rank. It receives neither the recorded reactive winner nor any temporal, source, or future fact. The shared score is masked by the unchanged shield and converted to a categorical distribution.

The primary gate is now the proper score available from capture: the upper endpoint of a 2,000-draw whole-episode bootstrap interval for L1e minus frozen L1d cross-entropy to the exact recorded behavior distribution must be below zero. To prevent a tiny relative gain from promoting a poor model, validation $\mathrm{KL}(\mu\|\pi)$ must also be at most 0.1 nat. The diagnosed structure must be repaired directly: validation argmax agreement with the reactive action must be at least 0.95, and pooled agreement on focused- or lexical-decisive rows must be at least 0.5. All thresholds are frozen before fitting. The hard training target remains unchanged so this is one structural ablation; exact propensities are used for audit and evaluation only. L1e runs no Wine and can at most admit a separately preregistered integration canary.

L1e ran at solver commit `ec24e8ad05a6e25513ad0468e22bb53617ebc338`. It exhausted all 10,000 updates at final-to-initial train-gradient ratio 0.011684, above the frozen 0.01 threshold, so its immutable decision is `inconclusive-l1e-shared-action-optimization-not-converged`. Validation does not override that stopping decision. Reportable validation evidence was also uniformly unfavorable: $\mathrm{KL}(\mu \| \pi)$ was 0.986686 versus 0.563604 for L1d; the complete-episode L1e-minus-L1d exact-target cross-entropy interval was $[0.408315, 0.437710]$; reactive-action agreement was 0.655371 versus 0.748550; and focused/lexical final-tie agreement was exactly zero. Train KL 0.978077 and reactive agreement 0.660861 show the same pattern, so episode-split overfit is not a sufficient explanation. The plan, model, and result SHA-256 digests are respectively `c4c2ef915c15 8544f73ddc779acb2e28697c20e4e587081153d13421bc22b60d`, `8afc439fc6678a088fd005308645 0d64f8344dc559382da116404759df5501f9`, and `a77b661136e3db85face5661c87919a13cf7966900 b7b3fbd6a48cea6e2edce9`. No Wine canary ran.

The fitted normalized weights further localize the practical failure. The clearance and boundary-reserve weights were positive, 5.611145 and 8.651481, but stationary and focused weights were negative, -0.288433 and -0.633078, and lexical priority was nearly zero and negative. Those signs oppose the collector precisely inside its rare final tie strata. A global additive score cannot turn lower-priority fields off when a higher-priority field differs; their correlations over common non-tie rows compete with their conditional meaning. Thus sharing action weights alone does not implement lexicographic, non-compensatory semantics. Because the fit did not converge, L1e does not formally reject every shared linear fit; its wide performance deficit does reject this frozen run as a candidate and does not justify another post-hoc timebox extension.

More fundamentally, exact capture already supplies the behavior distribution, and portable current features already reconstruct its reactive winner on every row. Generic-network imitation of that known hand-coded controller is a model-class probe, not a necessary global prerequisite for learning factual HIT risk. The next bounded work therefore returns to the original diagnostic sequence: current-observation action-conditioned execution/dynamics probes, then observed-shield-collapse and fixed-horizon HIT probes. A fixed short history is admitted only if those expose temporal ambiguity. This reframing does not admit IQL or revise any L1–L1e decision.

9.5 E4: minimal offline policy improvement

E4 begins only after E3 identifies one frozen portable representation. It compares BC with one offline expected-HIT method, initially IQL, on the same decision rows. Offline validation reports proper cost-prediction scores, calibration, action support, and episode-bootstrap intervals. Promotion then requires normal-speed original-Wine complete routes, zero Bomb, passing infrastructure contracts, lower HIT count under a preregistered serial comparison, and reported NMNB rate. Exact paired claims are allowed only if a separate exact-prefix replay gate passes; otherwise independent complete-route episode intervals are used. No offline estimate alone can promote a scorer.

9.6 E5: collection equivalence

Parallel workers are enabled only after E0. Serial and parallel fixed-seed runs must first agree on factual/action traces before independent natural-RNG collection begins. The first declared collection policy is an immutable 20% mixture of uniform exploration over the observed-shield set and the generic reactive baseline, with complete propensities and independent policy RNG streams per episode. Clock acceleration is a distinct hypothesis and cannot produce evidence until it passes an unchanged-retail-clock differential contract.

At commit `76782a4f37e0d12a9a2384561b53e68ceaf998ae`, the first declared shared-host test used two workers with eight logical CPUs each, disjoint game/controller partitions, retail frame multiplier 1.0 with coherent capture suspension, fixed retail seed `0x1234`, and the immutable 20% uniform-shield policy. The outer process tree used `SCHED_OTHER` nice 19 and idle I/O priority so normal-priority background proof jobs retained scheduling precedence. The exact command was:

```
ionice -c 3 nice -n 19 taskset -c 32-47 \
  .venv/bin/python scripts/gate_parallel_wine.py \
  --pool reference/wine-workers-v2-zklean-safe/pool.json \
  --policy-plugin src/th06_rl/policies/uniform_shield_exploration.py \
  --policy-state config/uniform_shield_exploration.json \
  --practice-stage 4 --diagnostic-rng-seed 0x1234 \
  --artifact-root artifacts/parallel-gate-exp02-76782a4-zklean-safe-nice19 \
  --corpus-root corpora/parallel-gate-exp02-76782a4-zklean-safe-nice19 \
  --output artifacts/parallel-gate-exp02-76782a4-zklean-safe-nice19/gate.json
```

The retail executable, native kernel, policy plugin, policy state, and worker pool SHA-256 digests were respectively `9f76483c46256804792399296619c1274363c31cd8f1775fafb55106fb852245`, `d26439f01400121a99137ff7d411b4bb6a2ff480daaaf1195e85a1b6cb671673`, `c70a2ad0a03bb189a01baabb62f6f404f6c4c91aa0e16c86645bfc0925f3e805`, `a63fbbb0b6d5886d327eb28b4b3d683cc13c6c759bffde8f9b2f554649068ef1`, and `76fe8c634855a2a9f11f4a5b83ab0c3498eb914df384ed40f206f199750447ad`. The pool and audit schemas were `th06-rl-normal-speed-wine-pool-v2` and `th06-rl-infra-audit-v2`.

Run	Frames	Transitions	HIT	Gaps (%)	Successors	Shield
Serial 0	25,106	25,105	16	0.0159	24,125	62
Concurrent 0	23,154	23,153	10	0.0346	22,535	64
Concurrent 1	25,455	25,454	13	0.1729	24,617	64

Table 5: E5 fixed-seed differential. Every row individually completed with zero Bomb, successor mismatch, unsafe divergence, conservative divergence, drop, callback failure, integrity error, or leftover process.

The run IDs were `20260821T101125Z-298490100`, `20260821T102110Z-520526700`, and `20260821T102110Z-450384400`; their audit SHA-256 digests were `5393077138d1a769e12a703e35e3cae11201e79859529acfcf52b40231153366`, `8a07cabb6ca93b897853c79ede1fc7cd24151f0a356b8390981cb2d498ba8aaf`, and `8bfaff06bc25ec808f258cc0e5d1b576627741e5b53928461b639fdd43035e4d`. Capture p99 was 8.420, 8.340, and 8.692 ms; solve p99 was 0.815, 0.784, and 0.895 ms. Thus every run passed its individual latency and semantic checks, but both HIT count and normalized factual digest failed exact equality. The gate rejected `concurrent-worker-0` and emitted no admission ledger. This is a negative E5 result, not permission to weaken equality: parallel collection remains disabled.

An earlier `SCHED_IDLE` attempt is retained as a scheduling negative: its serial run completed with zero semantic mismatch but 883 observation gaps (3.62%), above the unchanged 0.5% limit. Its audit digest is `00a5a1f8fab26fac909c4f0d421ba9ece13c418b25bc81ccdefdfc9c2a68da12`.

9.7 E6: frozen-batch interaction

E6 is admitted only if E4 records a specific coverage failure or an improved candidate whose remaining error is support-limited. One immutable policy is used for each complete Wine run; parameters never update or reload within a run. The first additional collection comparator is the declared 20% uniform observed-shield mixture. Any uncertainty- or novelty-guided collector is a separate later arm with a minimum action-probability floor and the identical Wine interaction budget. All episodes remain complete, immutable, and linked to their behavior distributions. E6 was not run.

9.8 E7: transfer

E7 replaces only the environment adapter and configuration and reports unchanged versus changed code/schema surfaces, zero-shot behavior, and fine-tuning sample cost. It was not run.

This documentation checkpoint follows E0, completed E2, and negative E3 and E5 gates. E3 collected its complete serial Stage 4 inventory and produced the initial fit plus two optimization-only follow-ups. The initial model improved held-out likelihood but failed calibration and was never executed online. The bounded diagnosis is complete; offline-only L1b improved held-out metrics but did not converge within its frozen timebox and is inconclusive. L1c then converged and failed the unchanged calibration gate. The read-only residual diagnosis selected one small current-observation MLP without reading scaled validation. That model converged and improved held-out proper scores but failed calibration. The completed target-contract diagnosis found exact propensities and localized the residual to flat-model misspecification after its other frozen explanations missed their materiality thresholds. The selected shared-action scorer then failed to converge and was worse than L1d on both splits, exposing additive-score mismatch with the lexicographic rule. Current-observation execution/dynamics and factual-risk probes are next; there is no authorized learned candidate.

10 Threats to Validity

The shield horizon may be too short, and unknown births may violate its narrow claim. Expected HIT count is only a surrogate for NMNB probability and may rank nonzero-HIT policies differently. HIT continuation changes later physical occupancy through lifecycle, resources, rank, bullet clearing, and invulnerability; complete-route training and survival-conditioned diagnostics must therefore be reported separately. Behavior action support does not guarantee support for histories reached by a stronger no-HIT policy, and a positive propensity floor does not prevent long-horizon OPE variance.

Complete-route episodes may be too few for stable episode-level intervals, future-HIT prediction, or uncertainty estimates. Wine scheduling may introduce observation gaps; command pickup may be mistaken for a new policy decision; and Touhou 6–specific memory facts may leak through an apparently generic feature builder. E0 and E2 are therefore prerequisites, not housekeeping. The transfer claim is falsified if downstream learner, corpus, objective, or evaluator semantics require title-specific branches.

11 Artifact and Reproducibility Contract

The implementation uses repository-relative scripts. The LaTeX artifact is built by `scripts/check_paper.sh`. Runtime setup, native compilation, Wine execution, corpus validation, and route verification have distinct entry points. The canonical compiled paper is tracked at `paper/main.pdf`; intermediate LaTeX files stay in the ignored `paper/build/` directory. Wine prefixes, corpora, and fitted models remain untracked outputs. Committed experiment claims must point to immutable hashes rather than mutable filenames.

12 Conclusion

The project asks how much complete-route HIT reduction can be learned from portable physical history when exact Wine interaction is costly, input pickup is delayed, future births are partially observed, and the online filter covers only instantiated hazards. The goal is NMNB; the first optimization claim is strictly expected undiscounted HIT reduction. The initial learner has three parts: an audited decision-epoch view, transparent BC, and immutable shield-constrained Wine evaluation. Every additional representation, algorithm, objective, uncertainty mechanism, or collection strategy must solve one measured failure of that baseline.

E0 establishes the serial factual data plane, E2 establishes the causal learner projection and first exporter, and E5 rejects the current parallel pool. E3 completed its serial Stage 4 inventory and initial frozen supervised fit; held-out likelihood improved but calibration failed, so it emitted no online candidate. The diagnosis localized the next test to convergence of that same transparent fit. L1b improved held-out likelihood and calibration without a representation change but exhausted its optimizer timebox before the frozen gradient gate. L1c changed only that ceiling, reached the train-only gate, and still failed held-out calibration. The linear current-observation candidate is therefore rejected under the joint gate. Train-only scale correction was insufficient, current features exactly reconstructed the collector, and the fixed small MLP then improved held-out NLL but also failed calibration. The frozen target-contract diagnosis verified every recorded propensity, found the sampled-label, stopping, and discarded-soft-target explanations insufficient, and localized the residual to the flat model's handling of a reconstructible action-conditioned lexicographic rule. The selected shared additive scorer then exhausted its timebox and performed worse, with zero agreement on the final tie strata. The known collector distribution remains an exact data-plane control; the next learner evidence is current-observation execution/dynamics and factual HIT-risk probes, with history admitted only by measured temporal ambiguity. No learned policy has run online, exact-prefix branching is not established, and this checkpoint makes no policy-improvement, NMNB, or transfer claim.

References

[1] Lili Chen, Kevin Lu, Aravind Rajeswaran, Kimin Lee, Aditya Grover, Michael Laskin, Pieter Abbeel, Aravind Srinivas, and Igor Mordatch. Decision transformer: Reinforcement learning via sequence modeling. In *Advances in Neural Information Processing Systems*, 2021. URL `https://arxiv.org/abs/2106.01345`.

[2] Ilya Kostrikov, Ashvin Nair, and Sergey Levine. Offline reinforcement learning with implicit q-learning. In *International Conference on Learning Representations*, 2022. URL `https://arxiv.org/abs/2110.06169`.

[3] Aviral Kumar, Aurick Zhou, George Tucker, and Sergey Levine. Conservative q-learning for offline reinforcement learning. In *Advances in Neural Information Processing Systems*, 2020. URL `https://arxiv.org/abs/2006.04779`.

[4] Manzil Zaheer, Satwik Kottur, Siamak Ravanbakhsh, Barnabás Póczos, Ruslan Salakhutdinov, and Alexander J. Smola. Deep sets. In *Advances in Neural Information Processing Systems*, 2017. URL `https://arxiv.org/abs/1703.06114`.